Protocol for Treating Musculoskeletal Disorders (MSDs) Using Low-Intensity Shockwaves
Device: MoreNova DT, MoreNova Pro with C30 and C15 Transducers.
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1. Purpose and Scope

This protocol provides a standardized treatment plan for using low-intensity shockwaves to manage musculoskeletal disorders (MSDs), including chronic pain, tendinopathies, myofascial trigger points, and musculoskeletal injuries. It is intended for healthcare providers using the MoreNova Device with the C30/C15 Transducers to target affected muscle and tendon regions. This protocol applies to patients aged 18–80 years diagnosed with MSDs, including but not limited to:

- Tendinitis (e.g., Achilles tendinitis, rotator cuff tendinitis)
- Myofascial pain syndrome
- Chronic lower back pain
- Osteoarthritis-related pain
- Plantar fasciitis
- Post-injury rehabilitation

2. Introduction

MSDs are among the leading causes of disability and pain worldwide. Traditional treatments such as physical therapy, medications, and surgical interventions may have limitations. Low-Intensity Extracorporeal Shockwave Therapy (LiESWT) provides a non-invasive approach that enhances healing by promoting angiogenesis, reducing inflammation, and stimulating cellular regeneration in affected tissues.

3. Scientific Review

Current Treatments:

- Drug therapy: NSAIDs and corticosteroid injections provide symptomatic relief but do not address underlying causes.
- Physical therapy: Effective for muscle strengthening and mobility but may have limited impact on chronic conditions.
- Surgery: Reserved for severe cases but involves longer recovery and potential complications.

Advantages of LiESWT:

- Non-invasive and painless procedure with no significant complications.
- Promotes angiogenesis and enhances tissue repair.
- Reduces musculoskeletal pain by improving microcirculation and modulating inflammatory responses.
- Effective in treating chronic conditions resistant to conventional therapies.

4. Patient Selection Criteria

Inclusion Criteria:

1. Adults (18–80 years) diagnosed with MSDs.
2. Persistent symptoms for at least 3 months despite conservative treatment.
3. Patients with tendinopathies, myofascial pain, or post-injury musculoskeletal pain.

Exclusion Criteria:

1. Implants and Electronic Devices

Active Electronic Implants: Pacemakers, Cardioverters, Defibrillators, or Neurostimulators.

Metallic Implants: Any metallic or orthopedic implants (screws, needles, etc.) in or near the treatment area.

2. Malignancy and Cancer History

Active Cancer: Known or suspected malignancy at or near the treatment site.

History of Cancer: Patients in recovery (typically up to 1 year post-treatment) due to the risk of stimulated angiogenesis (VEGF) affecting dormant cells.

3. Blood and Vascular Disorders

Bleeding Disorders: Clinically significant chronic hematological diseases.

Anticoagulants: Patients on blood thinners (e.g., Coumadin) with an INR > 3 (risk of hematoma).

Severe Neuropathy: Severe peripheral neuropathy with loss of sensation or polyneuropathies with hypersensitivity to pain.

4. Vulnerable Organs and Tissues

Gas-filled Organs: Direct treatment over lungs or bowel.

Central Nervous System: Brain, spinal cord, and major nerve clusters (including the skull and

spinal column).

Radiotherapy Area: Tissues previously treated with radiation (due to cellular damage and poor healing response).

Active Infection: Areas with cellulitis, abscess, or osteomyelitis.

5. Musculoskeletal and Bone Health

Growth Plates: Children in growth phases (risk of Epiphysiolysis).

Bone Integrity: Acute fractures or severe osteoporosis (risk of microfractures) in the treatment area.

Advanced Degeneration: Gross osteoarthritic changes in the focus area.

6. Pregnancy or lactation.

7. General Factors

Psychological Pathology: Psychiatric conditions that may hinder treatment compliance or safety.

5. Pre-Treatment Screening

To determine patient suitability for LiESWT, conduct the following:

- **Comprehensive medical history review** to assess contraindications.
- **Physical examination** with palpation of affected areas to determine treatment sites.
- **Imaging (X-ray/MRI/Ultrasound)** if needed to assess tissue integrity.
- **Pain assessment** using the Visual Analog Scale (VAS) or Numeric Rating Scale (NRS).

6. Treatment Protocol

Preparation:

- Conduct a thorough medical history review and physical examination.
- Educate the patient on procedure benefits, risks, and expected outcomes.

Procedure:

- **Number of Sessions:** 3-6 sessions depending on severity.
- **Session Frequency:** Twice per week.
- **Session Duration:** 15–20 minutes.
- **Applications Per Session:** 2–4, depending on pain localization.
- **Shocks Per Application:** 500-750 .

Treatment Parameters:

- **Energy (HV) Settings:** 1.6-1.9 kV.

- **Frequency (FRQ):** 2 Hz.

Post-Treatment Care:

- Advise patients to avoid strenuous activity for 24 hours.
- Encourage hydration and mild stretching to support recovery.
- Recommend continued physiotherapy if applicable.

7. Treatment Areas and Techniques

The treatment focuses on:

- **Tendinopathies:** Achilles tendon, rotator cuff, lateral epicondyle (tennis elbow).
- **Myofascial trigger points:** Commonly in the trapezius, quadratus lumborum, and gluteal muscles.
- **Joint pain:** Knee osteoarthritis, hip bursitis, and plantar fasciitis.

Transducer Placement:

- The first application should be performed at the source of the pain. Subsequent applications are recommended at 1-cm intervals along the tissue, to the right and left.
- Apply ultrasound gel for optimal transmission.
- The transducer is held firmly against the skin.
- Administer shocks in a systematic manner across the treatment zone.

8. Patient Comfort and Safety

To ensure patient comfort and safety:

- Position the patient to allow access to the treatment area while minimizing discomfort.
- Use conductive gel to improve shockwave penetration and reduce irritation.
- Monitor patient feedback and adjust settings as needed.

9. Treatment Outcomes and Follow-Up

Expected Results:

- Reduction in pain intensity after 1–4 sessions.
- Improved functional mobility and reduced stiffness.
- Enhanced healing of musculoskeletal structures.

Follow-Up Visits:

- Schedule assessments at 4 weeks and 12 weeks post-treatment.
- Evaluate pain levels using the VAS or NRS and assess functional improvements.
- Consider maintenance therapy every 3–6 months for chronic conditions.

10. Visuals and Diagrams

- **C30/C15 Transducers Placement:** Recommended positioning for tendinopathies, myofascial pain, and joint conditions.

- **Shockwave Application Patterns:** Distribution techniques for different MSDs.

11. Quality Control

Device Calibration:

- Verify device settings before each session.
- Ensure MoreNova Device recognition of preprogrammed parameters.

Clinician Training:

- All operators must be trained in MoreNova Device operation and C30/C15 Transducer usage.
- Clinicians should review the "Instructions for Use" provided by the manufacturer.

Checklist for Each Session:

- Confirm patient eligibility and absence of contraindications.
- Ensure the device and transducer are clean, sanitized, and fully functional.
- Verify correct transducer placement based on anatomical targeting.

12. References

For clinical studies and additional resources, visit the Initia medical Clinical Data Center.